

I- Choose the correct answer:

1-All of the following are essential structures of bacteria except:

- a- Cell wall b- Cytoplasm
- c- Spores d- DNA

2- The function of cytoplasmic membrane is:

- a- Shape of bacteria b- Motility
- c- Selective permeability d- Resist dryness

3-The structure that is formed under unfavorable condition is:

- a- Pili b-Capsule
- c- Spores d-Flagella

4-The function of flagella is:

- a- Respiration **b-Movement**
- c- Reproduction d) Fermentation_

5- Bacteria have very small size measured by:

- a- Microns b- Nanometers
- c- Millimeter d- None of above

6-The structure that is responsible for the shape of bacteria is:

- a- Cell wall b-Spores
- c- Cytoplasm d- Capsule

7- Bacteria need all of the following for their growth except:

- a- Nutrition b-Gases
- c-Light** d-Moisture

8-Which of the following cell can kill tumor cell directly:

- a. T helper
- b. T cytotoxic**
- c. Plasma cell
- d. Phagocytic cell

9-The following are characteristics of adaptive immunity EXCEPT:

- a. Specific.
- b. Acquired due to previous exposure.
- c. Depends on lymphocytes which have specific receptors.
- d. **Do not have memory.**

10-Antibodies are secreted by:

- a. T helper lymphocytes.
- b. **Plasma cells.**
- c. Cytotoxic T lymphocytes.
- d. Phagocytes.

11- optimum temperature for most of the pathogenic bacteria is

- a -47 °c b- 37 °c c -27 °c d- none of these

12-Bacteria which can grow in absence of oxygen are known as

- a-strict aerobes b-anaerobes
c-microaerophilic d-none of these

13- Which of the following bacteria is comma shaped

- a- Streptococci b- vibrio Cholera
c- Diphtheria d- none of these.

14-Sterilization by hot air oven requires temperature at

- a- 160 °c for 60 minutes b-121 °c for 15 minutes
C-100 °c for 120 minutes d- 80 °c degrees for 30 minutes

15-Sterilization by autoclave is carried out at

- a- 121 °c for 15 to 20 minutes
b-80 °c degrees for 30 minutes
C- 100 °c for 30 minutes
d-160 °c for 60 minutes

16-Moist heat under pressure is used in

- a-Tyndallization b-Inspissation
c-Pasteurization d-autoclave

17- Seitz filter is sterilized by

- a-Hot air oven **b- Autoclave**
b- Ultraviolet rays d- Ethylene oxide

18-Infrared sterilizer is a method of

- a- Heat sterilization
- b- Chemical sterilization
- c- Sterilization by radiation
- d- Sterilization by filtration

19-Seitz filter is used

- a- to obtain bacteria free filtrate
- b- to sterilize antibiotic solution
- c- to sterilize serum
- d- all of these**

20- dry heat sterilization is done by

- a-autoclave
c- incubator
- b- hot air oven**
d- boiling

21-Pasteurization is done at

- a- 63 for 30 minutes
b- 72 for 15-20seconds
c-both of these
d-none of these

22-Bacteriostatic agent means

- a- kills bacteria
- b- inhibit growth of bacteria**
- c- stimulate growth of bacteria
- d- none of these

23-Bacteriocidal agent means

- a-inhibit growth of bacteria
- c- stimulate growth of bacteria
- d- kills bacteria**
- e- all of these

24-The preferred vaccine for diphtheria consists of

a- Heat killed *corynebacterium diphtheria*

b-Attenuated *c.diphtheria*

c- Capsular polysaccharide

d- Toxoids

25- BCG is used for

a- vaccination for tuberculosis

b- vaccination for diphtheria

c- vaccination for Cholera

d- none of these

26- IgM is characterized by

a- produced in secondary immune response

c-produced in primary immune response

b-cross placenta

d-none of these

27- IgG is characterized by

a-produced in primary immune response

b-cross placenta

c-present an allergic reaction

d-none of these

28-Food poisoning with 1-6 hours incubation periods is due to

a- *salmonella enteritidis*

b- *Staphylococcus aureus*

c- *Clostridium botulinum*

d- none of these

29-Food poisoning caused by toxin is due to

a- *cl.botulinum*

b- *salmonella enteritidis*

c- *Staphylococcus aureus*

d- A&B

e- A&C

30-Epidemic cerebrospinal meningitis is caused by

a-*Neisseria meningitidis*

b- *Neisseria gonorrhea*

- c- *Salmonella*
- d- none of these

31-Symptoms of Clostridium food poisoning are

- a- vomiting and diarrhea
- b- respiratory paralysis**
- c- fever
- d- none of these

32- Enteric fever is caused by

- a- *Staphylococcus aureus*
- b- *streptococcus pyogens*
- c- *salmonella typhi***
- d- *vibrio Cholera*

33-Tuberculosis is transmitted by

- a-droplet spread**
- b-through skin and mucous membrane
- c-sexual route
- d-none of these

34-IgE is characterized by

- a- present in allergic reaction**
- b- cross placenta
- c-produced in primary immune response
- d-none of these

35- Transmission of nosocomial infection by

- a- Contact
- b- Airborne
- c- Contaminated vehicles
- d- All the above**

36- Prevention of nosocomial by :

- a- Frequent hand washing for all hospital persons
- b- Proper sterilization and disinfection .
- c- Education of hospital staff.

d- All the above

37-The most likely agents to cause a nosocomial infection include:

a-staphylococci and pseudomonas

b-Parvovirus

c-Streptococci

d-Markes disease

38-An infection that is acquired in the hospital is calledinfection:

a- Parasite

b- Pathogen

c- Nosocomial

d- Anaerobic

39-Factors influencing the nosocomial infection are

a- Patient susceptibility

b- Microbial agent

c- Environment factors

d- All the above

40-Which of the following is the first priority in preventing infections when providing care for a client.

a- Hand washing

b- Wearing gloves.

c- Using a barrier between client's furniture and nurse's bag.

d-Wearing gowns and goggles.

41- Patients who have a suspected bacterial nosocomial infection:

a- acquired the infection while in hospital.

b- are likely to require antibiotic therapy.

c- should be treated as infectious until confirmed otherwise.

d- All of the above.

42- A procedure which often predisposes a hospitalised patient to a urinary tract infection is:

a- urinary catheterisation.

b- antibiotic therapy.

c- colonisation of periurethral areas with 'hospital organisms'.

d- immunosuppressive therapy.

43- Which of the following is a common type of hospital acquired infection?

a- Urinary tract infection

b- Surgical wound infection

c-Lower respiratory tract infection

d-All of the above

44-An example of an endogenous infection would be:

a- infection of a surgical wound with organisms from another patient.

b- lung infection in which the causative organisms are inhaled.

c- cystitis caused by organisms from the person's own gastrointestinal tract.

d- surgical site infection with MRSA.

45-Which of the following would not usually be considered the source of an exogenous nosocomial infection?

a-other patients

b-patient's normal flora

c-IV cannula

d-hands of staff

46-Phagocytosis is one of the function of:

a) Macrophages

b) NK cells

c) T helper cells

d) T cytotoxic cells

47-The antibodies that are produced in the primary immune response are:

a) IgG

b) IgM

c) IgE

d) IgD

48-Antibodies that can cross the placenta are:

- a) IgG
- b) IgM
- c) IgE
- d) IgD

49-Antibodies that are present on mucous membranes are:

- a) IgG
- b) IgM
- c) IgE
- d) IgA

50-Antibodies that are present in Type I hypersensitivity are:

- a) IgG
- b) IgM
- c) IgE
- d) IgD

51-Incompatible blood transfusion is an example of:

Type I: Immediate hypersensitivity or anaphylactic shock:

- a) **Type II: Cytotoxic reactions:**
- b) Type III: Immune complex reactions:
- c) Type IV: Delayed hypersensitivity

52-Injection of penicillin may lead to anaphylactic shock which is an example of:

- a) **Type I**
- b) Type II
- c) Type III
- d) Type IV

53- In case of respiratory tract infection. The following sample should be examined:

- a) Urine
- b) Stool
- c) **Sputum**
- d) Pus

54-In examination of urine sample. The following precautions are essential:

- a) Morning sample
- b) Clean skin around urethra
- c) Mid-stream
- d) All of the above**

55-Selective toxicity of antibacterial agents means that:

- a) The antibiotic kills all bacteria
- b) The antibiotics kills host cells
- c) The antibiotic is broad spectrum
- d) The antibiotic is harmful to the microbe but harmless to host cells**

56-Complications of the chemotherapy include

- a) Toxic side effects
- b) Resistance to drugs
- c) Hypersensitivity reactions
- d) All of the above**

57-Streptococcus pyogens causes:

- a) Sore throat**
- b) Pneumonia
- c) Tuberculosis
- d) Diarrhea

58-The organism that causes diphtheria produces

- a) Toxin**
- b) Enzyme
- c) Vitamin
- d) Growth factor

59-The test for diagnosis of rheumatic fever is:

- a) Oxidase

- b) Catalase
- c) Coagulase
- d) ASOT**

60-The vaccine for prophylaxis of diphtheria is prepared from

- a) Capsule
- b) Dead bacteria
- c) Toxoid**
- d) Living attenuated vaccine

61-Pseudomembrane formation on the tonsils is caused by

- a) Streptococci
- b) Neisseria
- c) Diphtheria**
- d) None of the above

62-Complication of streptococci include

- a) Rheumatic fever
- b) Glomerulonephritis
- c) Urinary tract infection
- d) Both A and B**

63-Whooping cough is caused by:

- a) Bordetella**
- b) Streptococci
- c) Tuberculosis
- d) None of the above

64- Swabs are appropriate for specimens collected from all the following sites EXCEPT:

- A) nasal

B) ear

C) Urine

D) nasopharynx

65--urine sample used for a urine culture is

A) the first portion of the urine

B) mid-stream urine

C) last portion of urine

D) whole urine collected

66-important precautions should be taken in collecting clinical specimens includes a label firmly attached to the container with the following information

A) sex

B) Name

C) age

D) all of the above

67-When instructing the patient to collect a sputum specimen, the medical assistant should tell the patient to

A) best to collect sputum specimens in the evening

B) collect sputum from a first morning coughing episode.

C) allow saliva and mouth mucus to enter cup

D) sputum culture can be done from saliva

68-When instructing the patient to collect a stool specimen, the medical assistant should tell the patient to..

A) Place something in the toilet to catch the stool

B) Place urine in the cup also

C) No problem that stool touch the inside of the toilet

D) Not necessary to urinate first

69-When you come to collect a blood culture specimen

- A) skin must then be decontaminated with a 10% alcohol
- B) Use blood culture bottles aerobic only
- C) The specimen should preferably be taken during fever episodes**
- D) re-palpate vein after decontamination.

70-When you come to collect a wound swab

- A) Obtain the specimen after use of antiseptic
- B) Usually very small amount of pus is sufficient
- C) Obtain the specimen prior to any dressing or cleaning procedure**
- D) It is better to have the sample after starting antibiotic therapy

71-Factors determine the effectiveness of antimicrobial agents

- A) The drug should have selective toxicity**
- B) The host should become allergic
- C) The organism should be resistant to the drug

72- Broad spectrum antibiotics

- A) act on gram +ve and gram -ve bacteria**
- B) act on gram +ve only
- C) act on gram -ve bacteria only
- D) as macrolides

73- causes of development of drug resistance

- A) adequate dosage,
- B) suitable duration of treatment,
- C) presence of infection
- D) the abuse of antibiotics without in vitro susceptibility testing**

74- one of causes of drug toxicity.

A) prolonged use of drug with narrow margin of selective toxicity.

B) Small dose of the drug

C) Use of broad spectrum antibiotics

D) The use of the same antibiotic more than one time

75- To limit drug resistance

A) use of drug after sensitivity test

B) administration of one drug in mixed infections

C) the use of valuable drug as vancomycin

D) not follow proper dose or duration

76- throat swab should be taken from

A) nasopharynx

B) nostril

C) the posterior pharynx and tonsillar area

D) cheeks, teeth or gums

**77- when collecting sputum specimens, health professionals should use
..... as droplets and aerosols may be generated**

A) full PPE

B) Mask only

C) Gloves only

D) No thing

78-What is wrong about blood culture sampling

A) skin must then be decontaminated with a 70% alcohol

B) Do not re-palpate vein after decontamination

C) Use aerobic blood culture bottle only

D) discard the needle in safety box.

79-All of the following are essential structures EXCEPT:

- a) **Capsule**
- b) Cell wall
- c) Bacterial nucleoid
- d) Cytoplasm
- e) Cytoplasmic membrane

80-All of the following may be the shape of bacteria EXCEPT:

- a) Cocci
- b) Bacilli
- c) Spiral
- d) **Square**
- e) Vibrio

81-..... is the structure responsible for motility:

- a) Pili
- b) Spores
- c) **Flagella**
- d) Cell wall
- e) Capsule

82-Which structure is responsible for resistance to heat and radiation in spores?

- a) **Keratin coat**
- b) Cytoplasmic membrane
- c) Cell wall
- d) Ribosomes
- e) Bacterial DNA

83-The viruses possess:

- a) **Only one type of nucleic acid DNA or RNA**
- b) Both DNA and RNA
- c) Neither DNA nor RNA
- d) Proteins only

84-The causative organism of gonorrhea is:

- a) **Neisseria gonorrhea**
- b) Staphylococcus aureus
- c) Corynebacterium diphtheriae
- d) Mycobacterium tuberculosis

85-One of the following diseases is sexually transmitted:

- a) **Syphilis**
- b) Gas gangrene
- c) Diphtheria
- d) Tetanus

86-Which of the following methods is used for viewing spirochetes?

- a) Bright-field microscopy
- b) Phase-contrast microscopy
- c) **Dark-field microscopy**
- d) Electron microscopy

87-One of the followings is not a character of *Clostridium tetani*:

- a) Have spores
- b) Drum-stick appearance
- c) **They are aerobes.**
- d) They are producer of powerful exotoxin.

88-Rice water stool is characteristic in:

- a) Enteric fever .
- b) Botulism.
- c) **Cholera.**
- d) Tetanus

89-Neurological manifestations are present in:

- a) **Botulism.**
- b) Brucellosis.
- c) Cholera.
- d) Puerperal sepsis

90-Intermittent pyrexia is characteristic in:

- a) Cholera.
- b) **Brucellosis.**
- c) Botulism.
- d) Tetanus

91-Widal test is used for diagnosis of:

- a) Cholera.

- b) Brucellosis.
- c) **Enteric fever.**
- d) Tetanus

92-TAB vaccine is used for protection from :

- a) Cholera.
- b) Brucellosis.
- c) **Enteric fever.**
- d) Tetanus

93-All of the following are characters of Hepatitis virus (A) EXCEPT:

- a) Transmitted by the fecal-oral route
- b) Ingestion of contaminated food or water
- c) **Turn to chronic infection**
- d) No carriers

94-All of the following are characters of Hepatitis virus (E) EXCEPT:

- a) Fecal-oral transmission
- b) Water-borne infection
- c) High mortality rates among pregnant women.
- d) **There is a vaccine for prophylaxis**

95-All of the following are characters of Hepatitis virus (B) EXCEPT:

- a) **No vaccine for prophylaxis**
- b) Transmitted by blood, sexual intercourse and perinatally
- c) Chronic carriers
- d) Leads to cirrhosis of the liver and HCC

96-All of the following are characters of Hepatitis virus (D) EXCEPT:

- a) Co-infection with HBV
- b) **No chronic liver disease**
- c) Super-infection of an already HBV-infected individual.
- d) Prevented by HBV prophylaxis

97-All of the following are characters of Hepatitis virus (C) EXCEPT:

- a) post-transfusion hepatitis
- b) **Available active or passive immunization**
- c) Transmitted parenterally and perinatally
- d) Chronic active hepatitis

98- The organism causing pyogenic tonsillitis is

- a) *Staphylococcus aureus*
- b) *streptococcus pyogens*
- c) *salmonella typhi*
- d) *vibrio Cholera*

99-- The organism causing whooping cough is

- a) *Staphylococcus aureus*
- b) *streptococcus pyogens*
- c) *salmonella typhi*
- d) *bordetella pertussis*

100- Complications of streptococcal tonsillitis is

- a) **Rheumatic fever**
- b) Enteric fever.
- c) Botulism.
- d) None of the above.

II-Put true on the right sentences and false on wrong one:

- 1-Bacteria are unicellular organisms. (T)
- 2-Suitable temperature for bacterial growth is 37 (T)
- 3-Bacteria are easily seen by naked eye (F)
- 4-Acidity or alkalinity is suitable for bacterial growth (F)
- 5-Some bacteria have capsule (T)
- 6-All the bacteria can form spores (F)
- 7- Viruses can replicate inside the living and non-living cells. (F)
- 8- The viruses possess only one type of nucleic acid. (T)
- 9- Bacteria possess only one type of nucleic acid. (F)
- 10- Viruses are ultramicroscopic, measured by nanometer (nm). (T)

- 11-staphylococcal enterocolitis one of Complications of **prolonged** Antibacterial Chemotherapy (T)
- 12-suppression of normal flora by the antibiotic used and their replacement with drug resistant organisms which cause disease is called drug toxicity (F)
- 13-Simultaneous administration of more than one drug, delays the emergence of resistance (T)
- 14-The culture specimens should be collected before the initiation of antibiotic treatment (T)
- 15- In nasopharyngeal swab a dry swab is inserted 2 cm into the nostril (F)
- 16-In patients suspected to have pulmonary Tuberculosis one specimens is sufficient (F)
- 17- When inoculating the blood culture bottles the anaerobic culture bottle should be inoculated first and then the aerobic culture bottle (T)
- 18-The blood culture specimen should preferably be taken during fever episodes (T)
- 19- Passive Immunity is exposure to ready antibodies. (T)
- 20 -Humoral immunity is carried out by B-cell. (T)
- 21- Innate immunity is specific. (F)
- 22- Sulfa drugs can be sterilized by hot air oven (T)
- 23- Autoclave is a type of moist heating. (T)
- 24- Rubber catheters can be sterilized by ethylene oxide. (T)
- 25- Ultraviolet rays have high penetration power. (F)
- 26- Seitz filter can be sterilized by autoclave. (T)
- 27- Diphtheria vaccine is in the form of toxin. (F)
- 28- Whooping cough vaccine is living vaccine. (F).
- 29-Post streptococcal complications are caused by the organism itself (F).
- 30-Streptococcal pyogenes give beta haemolysis on blood agar(T).
- 31- Endogenous source of nosocomial infection is normal flora of the patients himself. (T)

32- Exogenous sources of nosocomial infection are from infected patients & carriers. (T)

33- Frequent hand washing for all hospital personnel is the most important step for prevention of nosocomial infection. (T)

34- Implementation of a comprehensive infection control program should include education and training of hospital staff. (T)

35- Proper sterilization and disinfection don't help in prevention of nosocomial infection. (F)

37- Isolation of infective patients is important procedure for prevention of nosocomial infection (T)

38- Conservative antibiotic use helps in limitation of nosocomial infection.(T)

39- Nosocomial infections are always preventable (F)

40- IgA is the prominent immunoglobulin in various secretions (T).

41- IgM is a predominant immunoglobulin in the secondary immune response (F).

42- IgA is associated with allergy (F).

43- IgG cross the placenta inhuman(T).

44- DpT is used for vaccination against Tuberculosis (F).

45- adaptive immune response has a rapid action (F).

46- Alcohol is used as a skin antiseptic(T).

47- BCG is a living attenuated vaccine for tuberculosis (T).

48- diphtheria is treated by anti toxin and antibiotics(T).

49- surgical instruments is sterilized by autoclave (T).

50- upper respiratory tract infection means bronchitis and pneumonia(F).

51- Rheumatic fever is a complication off streptococcal tonsillitis(T).

52- Widal test is a serological test for diagnosis of enteric fever(T).

53- Diagnosis of enteric fever is done by obtaining blood sample in the second week(F).

54-Botulism is a food poisoning caused by consuming canned food(T).

55-Poliomyelitis is a viral disease acquired through the GIT. (T)

- 56- Rabies is a bacterial disease transmitted by the bite of a rabid animal. (F)
- 57- Serum hepatitis disease could be transmitted from a mother to her baby.(T)
- 58- Yellow fever disease is caused by a bacterial microbe (F)
- 59- There is no long lasting immunity after catching influenza disease. (T)
- 60- Common cold is caused by the Influenza virus. (F)
- 61- Orchitis is a complication of the Mumps disease. (T)
- 62- Herpes zoster disease is a disease of children. (F)
- 63- Herpes zoster is due to reactivation of the virus lying dormant in the patient. (T)
- 64- Poliomyelitis disease affects the anterior horn cells of the spinal cord. (T)
- 65- Hepatitis A Virus is acquired through contaminated syringes. (F)
- 66- Hepatitis E can be more severe in pregnant women and immunocompromised patients. (T)
- 67- Hepatitis C Virus is a DNA virus. (F)
- 68- Herpes simplex virus may infect the conjunctiva or the cornea. (T)
- 69- Sabin vaccine is a killed virus vaccine. (F).
- 70- The size of bacteria is measured in nanometers while the size of viruses is measured in microns (F)
- 71- Saprophytic bacteria live on inanimate bodies while parasitic bacteria live on living bodies (T)
- 72- Gram positive bacteria cell wall contains only 5-10% peptidoglycan (F)
- 73- Obligate anaerobes can live in the presence of oxygen (F).
- 74- Gonorrhea and syphilis are sexually transmitted diseases (T)
- 75- Clostridium tetani are spore forming bacilli(T)
- 76- Treponema pallidum is stained easily by Gram stain(F)
- 77- Secondary stage of syphilis is manifested by chancre (F)
- 78- Neisseria gonorrhea is intracellular gram -ve diplococci (T)

- 79-Clostridia which cause gas gangrene produce gas (T)
- 80-Staphylococcal carriers are very common especially among hospital staff. (T)
- 81-Brucellosis is associated with animals (T)
- 82-TAB vaccine is heat killed vaccine for enteric fever prophylaxis. (T)
- 83-Enteric fever is caused by Salmonella typhimurium. (F)
- 84-Botulism is caused by contaminated canned food.(T)
- 85-Bacillary dysentery is caused by Shigella species. (T)
- 86-Hepatitis (A) virus can be transmitted by blood transfusion (F)
- 87-Hepatitis (c) virus leads to liver cirrhosis and HCC(T)
- 88-Hepatitis (D) virus can be transmitted as coinfection with hepatitis (E) virus(F)
- 89-There are vaccines for Hepatitis A and B viruses' prophylaxis(T)
- 90-There are vaccines for Hepatitis E and C viruses' prophylaxis(F)

"GOOD LUCK"